Ventilator Usage
AW-3
Effective September 22, 2021

FFEMS utilizes two models of ventilators. The PARA PAC which composed both the Medic and Transport Ventilator. (See SOP 600.15 for more details and the ZOLL Z vent (See SOP ).

The Ventilator is used at the discretion pf the paramedic in the prehospital environment which may include during a respiratory/cardiac arrest and during ROSC. It may be used with an IGEL or ETT.

The Ventilator may be used on an interfacility transport. The prescribing ventilator parameters shall be initially ordered by the sending physician on the Transferring Order Form. These may be adjusted as needed using this protocol or parameters set by the sending physician It is always advisable to have a respiratory therapist in attendance when completing an interfacility transport using a ventilator.

The PARA PAC ventilator uses SIMMV mode exclusively and can be used on patient 5Kg and above. The same circuitry is used. PEEP may be applied with external control devices. It is volume controlled.

The Zoll Ventilator is capable of CPAP, BIPAP, Pressure support, SIMV and AC Modes. It can be volume controlled or pressure controlled. The pediatric/adult patient circuits are intended for use when delivering tidal volume from 200 ml to Adult. The infant/pediatric patient circuit is intended for use when delivering tidal volume from 50 ml to 300 ml. It may be used on peds 5Kg to adult.

Refer to Airway AW-1 for airway management of the patient.

Refer to Post Intubation G-10 protocol for pain and sedation management of the ventilated patient.

At all times, SPO2 and waveform capnography are to be monitored as well as continuous ECG and vital sign monitoring.

***Cardio/Respiratory Arrest Usage:*** (See Default parameters for ZOLL at end of protocol)

1. Confirm Definitive airway placement
2. Set up ventilator and ensure its operation in accordance with SOP
   a. Rate: Adult 10 bpm 6 ml/Kg TV FIO2 100% PEEP 5 cm H20 Pressure if available
      Additional ZOLL settings: MODE AC
   b. Child 20bpm 6 ml/Kg TV FIO2 100% PEEP 4 cm H20 Pressure if available
      Additional ZOLL settings: MODE AC (V)
   c. Infant 30 bpm 6 ml/Kg TV FIO2 100% PEEP 4 cm H20 Pressure if available
      Additional ZOLL Setting: MODE AC (V)
3. Attach patient to ventilator. Assess breath sounds. Assess oxygenation and ventilation. Be alert for alarms. Trouble shoot accordingly. It is likely the high pressure alarm may sound during compressions. See SOP
4. If at any time an alarm cannot be reconciled utilizing the DOPE pneumonic, return to BVM.

***ROSC Usage of Ventilator:***

1. Confirm Definitive airway placement
2. Set up ventilator as in Cardio/Respiratory Arrest and ensure its operation in accordance with SOP
3. Attach patient to ventilator. Assess breath sounds. Assess oxygenation and ventilation. Be alert for alarms. Troubleshoot accordingly. See SOP

   Adjust ventilator Respiratory Rate to maintain a capnography between 35-45. Make adjustments incrementally by 2 breaths per minute waiting 5-10 minutes between adjustments.

   "AT MD DISCRETION" Ventilatory Rates

   **ADULT:** < 12 and > 25

   **CHILD:** <15 and > 30

   **INFANT:** < 20 and > 35

   **Use caution when attempting to lower a respiratory rate to increase ETCO2 in patients in shock or with acidosis- a "normal" ETCO2 may cause them to arrest. Lower the respiratory rate only after consultation with medical control or the transferring physician.**

   Adjust FIO2 to maintain a SPO2 between 94% and 99%. Make adjustments incrementally by 10 waiting 5-10 minutes between adjustments. ( The ParaPac ventilator will only provide 50% or 100% FIO2)

4. If at any time an alarm cannot be reconciled utilizing the DOPE pneumonic, return to BVM.

***Interfacility Usage of Ventilator:***

1. Confirm Definitive airway placement
2. Set up ventilator in accordance with Sending Physician Parameters and ensure its operation in accordance with SOP

- When performing a ventilator switch on any patient:
- **DO NOT ATTEMPT TO SWITCH VENTILATOR WITHOUT RESPIRATORY THERAPY IN THE ROOM AND READY.**
- Set ventilator parameters on the transferring ventilator per orders, start ventilator and pause it
- Have respiratory therapy pause the sending facility ventilator
- Clamp the ETT with a Kelly clamp
- Switch Ventilators--Place the transferring ventilator tubing on to the ETT. Unclamp and restart the ventilator.

3. Assess breath sounds. Assess oxygenation and ventilation. Be alert for alarms. Trouble shoot accordingly. See SOP
4. If at any time an alarm cannot be reconciled utilizing the DOPE pneumonic, return to BVM

Adjustment of ventilator setting should be *FIRST m*ade in consultation with Respiratory Therapist if in attendance.

Adjustment of ventilator settings may be *Secondly* made by the paramedic in accordance with written guidelines by the sending physician.

If neither of these are present, the patient should not be allowed to become hypoxic or hypercarbic, UNLESS PERMISSABLE HYPERCARBIA IS PART OF THE TREATMENT AS PRESCRIBED BY THE SENDING PHYSICIAN. In this case the paramedic may incrementally adjust the ventilator every 5-10 minutes for sustained hypoxia < 94% (or sustained SAO2 < SAO2 obtained on transfer) or a sustained hypercarbia > 45 accordingly.

***Persistent SAO2 < 94%(or persistent SAO2< SAO2 obtained on transfer):*** Adjust PEEP AND FIO2 together as noted in table every 5-10 minutes until SAO2 > 94% consistently(or above falling SAO2). Begin at currently prescribed FIO2/PEEP the vent is already set on and gradually adjust parameter to chart if not already there. Once at chart values increase both parameters as noted to achieve goal.

| FIO2 | 30% | 40% | 40% | 50% | 50% | 60% | 70% | 70% | 70% |
|---|---|---|---|---|---|---|---|---|---|
| PEEP | 5 | 5 | 8 | 8 | 10 | 10 | 10 | 12 | 14 |

| FIO2 | 80% | 90% | 90% | 90% | 100% | | | | |
|---|---|---|---|---|---|---|---|---|---|
| PEEP | 14 | 14 | 16 | 18 | 18-24 | | | | |

***Persistent Capnography End Tidal CO2 > 45 (Unless allowed by sending MD)***

Make adjustments incrementally by 2 breaths per minute waiting 5-10 minutes between adjustments.

"AT MD DISCRETION" Ventilatory Rates
**ADULT:** < 12 and > 25
**CHILD:** <15 and > 30
**INFANT:** < 20 and > 35

***Persistent new Capnography End Tidal CO2 < 35***

**Seek Medical Control Advice.**

**Use caution when attempting to lower a respiratory rate to increase ETCO2 in patients in shock or with acidosis- a "normal" ETCO2 may cause them to arrest. Lower the respiratory rate only after consultation with medical control or the transferring physician.**

ZOLL DEFAULT PARAMETERS **MUST BE ADJUSTED**



## Adult Default Parameter Setting Values

| Parameter | Default Setting Value |
|---|---|
| Mode | AC (V) |
| BPM | 12 |
| I:E | 1:3 |
| VT | 450 |
| PEEP | 5 |
| PIP Limit | 35 |
| $FIO_2$ | 21 |



## Pediatric Default Parameter Setting Values

| Parameter | Default Setting Values | |
|---|---|---|
| Mode | SIMV (P)* | AC(P) |
| BPM | 20 | 20 |
| Ti | 0.6 | 0.6 |
| PIP | 20 | 20 |
| PEEP | 4 | 4 |
| PIP Limit | 30 | 30 |
| $FIO_2$ | 21 | 21 |
| * Note: SIMV (V) and SIMV (P) are optional modes on the Z Vent. | | |

## ***Cardiac Arrest Settings***

Caution: The High Pressure may alarm due to compressions and may not be able to reconcile. Always utilize DOPE initially to correct problems.

Adult: Choose ADULT Default—Change to

| | |
|---|---|
| Mode | AC (V) |
| Rate | 10 |
| TV | 6 ml/KG |
| PEEP | 5 |
| FIO2 | 100% |
| All others OK | |

Pediatric Child > 1

| | |
|---|---|
| Mode | AC (V) |
| Rate | 20 |
| TV | 6 ml/KG |
| PEEP | 4 |
| FIO2 | 100% |
| Al others OK | |

Infant < 1 yr

| | |
|---|---|
| Mode | AC (V) |
| Rate | 30 |
| TV | 6 ml/KG |
| PEEP | 4 |
| FIO2 | 100% |
| Al others OK | |

## ***ROSC***

Maintain SAO2 at 94-99%  Adjust FIO2 if 100%. Make changes in increments of 10 every 5-10 minutes
Maintain Capnography at 35-45  Assess and treat hypocarbia as poor perfusion first especially if B/P is < 90 systolic.
Use caution when attempting to lower a respiratory rate to increase ETCO2 in patients in shock or with acidosis- a "normal" ETCO2 may cause them to arrest.  Lower the respiratory rate only after consultation with medical control or the transferring physician
Adjust ventilator Respiratory Rate to maintain a capnography between 35-45.  Make adjustments incrementally by 2 breaths per minute waiting 5-10 minutes between adjustments.

"AT MD DISCRETION" Ventilatory Rates

**ADULT:** < 12 and > 25

**CHILD:** <15 and > 30

**INFANT:** < 20 and > 35

Adult:  Choose ADULT Default—Change to

| | |
|---|---|
| Mode | AC  (V) |
| Rate | 12-14 |
| TV | 6 ml/KG |
| PEEP | 5 |
| FIO2 | 100% |
| All others OK | |

Pediatric Child > 1  Choose Pediatric  Change to

| | |
|---|---|
| Mode | AC  (V) |
| Rate | 20 |
| TV | 6 ml/KG |
| PEEP | 4 |
| FIO2 | 100% |
| Al others OK | |

Infant < 1 yr  Choose Pediatric  Change to

| | |
|---|---|
| Mode | AC  (V) |
| Rate | 30 |
| TV | 6 ml/KG |
| PEEP | 4 |
| FIO2 | 100% |
| Al others OK | |